METRIC

MIL-PRF-21858E
20 January 1996
SUPERSEDING
MIL-T-21858D
18 November 1976

PERFORMANCE SPECIFICATION

TRUCK, HAND, SHELF, STOCK PICKER (METRIC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for multiple shelf, four caster, 227 kilogram (kg) (500 pound (lb)) capacity hand trucks (hereinafter called "trucks").

1.2 Classification. Trucks are of the following types as specified:

Type I	- With pushbar.
Type II	- With pushbar and ladder.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3920

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in section 3 and section 4 of this specification, whether or not they are listed.

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issue of the documents cited in the solicitation (see 6.2).

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC. AGENT

National Motor Freight Classification.

(Applications for copies should be addressed to the American Trucking Association, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a first article sample shall be subjected to the first article inspection in accordance with 4.2.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

3.2 Materials. Materials used shall be in accordance with the manufacturer's materials specification for handtrucks. The materials shall be capable of meeting all of the operational and environmental requirements specified herein (see 4.5.1). Recovered materials shall be used to the maximum extent practicable. Asbestos, cadmium, and radioactive material will not be used in this item. Radioactive material is defined by 1) Title 10, Code of Federal Regulations, Part 40, and 2) Other radioactive material in which the radioactivity is greater than 0.002 microcuries per gram or 0.01 microcuries total activity for the item.

3.2.1 Fastening devices. All bolts, capscrews and nuts shall be provided with lockwashers or other locking devices and shall be tightly drawn. Hardware shall be treated or plated in accordance with commercial practice to prevent corrosion.

3.3 Design and construction. Trucks shall be fabricated to the manufacturer's design and construction for trucks except for the size and rated load capacity which shall be as specified in table I (see 4.5.2).

TABLE I. Physical characteristics of trucks.

Type	Rated capacity kg (lbs)	Minimum truck dimensions millimeter (mm) (inch (in))			Component
		Length	Width	Height	
I	227 (500)	1092 (43)	508 (20)	830 (32.7)	Frame (with push bar)
		997 (39.25)	489 (19.25)	38 (1.5)	Shelves (two stationary and one removable)
II	227 (500)	1092 (43)	508 (20)	830 (32.7)	Frame (with push bar and ladder)
		997 (39.25)	489 (19.25)	38 (1.5)	Shelves (two stationary and one removable)
			464 (18.25)	762 (30)	Ladder

3.3.1 Casters. The casters shall be securely attached to the underside of the frame, each located as close as possible to each of the four corners. The swivel casters shall be capable of fully swiveling with truck loaded to rated capacity. Caster size shall be fully compatible with the rated capacity of the truck.

3.3.2 Push bar handle. Each truck shall be provided with a push bar handle assembly. Variation in design of the push bar assembly is permissible provided the assembly is equivalent in strength, hand clearance and is of the bolted-on type. Handles shall be placed at protected places on the racks or truck body so that passing traffic, walls or other objects will not crush or scrape the operator's hands.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

3.3.3 Ladder. The Type II trucks shall also be provided with a folding ladder which shall be bolted on. Surfaces of the ladder steps shall be painted with a nonslip walkway compound containing 50% total solids weighing 678-848 gram per square meter (20-25 ounces per square yard). Variation in design to provide for a combination push bar and ladder are permitted provided all other requirements of 3.3 are met. Replacement tips of a nonslip composition shall be attached to the bottom of the ladder legs. The hind section of the ladder shall have a foot operated floor lock to prevent movement of the truck.

3.4 Performance.

3.4.1 Static load. The truck shall not be damaged when subjected to the static load specified in 4.4.2.2. The nonslip walkway compound applied to the surfaces of the ladder steps (see 3.3.3) shall conform to the minimum requirements in table II.

TABLE II. Performance requirements of nonslip walkway compound for ladder steps.

Minimum factor of sliding friction	Dry	Watered	Oiled
Leather	0.50	0.80	0.45
Rubber	0.90	0.80	0.60

3.4.2 Shock load. Truck shall not be damaged when subjected to shock load specified in 4.4.2.1.

3.5 Identification markings. The truck shall be identified by the attachment of a metal plate securely attached with screws, bolts or rivets, and be conspicuously located. The plate shall contain the manufacturer's name, CAGE and part number.

3.6 Treatment and painting. The frame and finished boards of the truck shall be cleaned, treated and painted in accordance with normal commercial practice to assure complete coverage and durability of the finish. Unless otherwise specified (see 6.2), color shall be gloss yellow.

3.7 Workmanship. All parts, components and assemblies of the truck including castings, forgings, molded parts, stampings, bearings, seals, machined surfaces and welds shall be clean and free from sand, dirt, fins, pits, cracks, sprues, scales, flux and other harmful extraneous materials and defects. External surfaces shall be free from burrs, sharp edges and corners except when sharp edges and corners are required.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of inspection. The inspection conditions specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. Unless otherwise specified (see 6.2), first article inspection shall be performed on preproduction or initial production samples as specified when a first article is required (see 3.1). This inspection shall include the examinations of 4.4 (see table III) and the tests of 4.4.2.1 through 4.4.2.2 (see table IV).

TABLE III. Classification of defects.

Category	Defect	Method of examination
Major		
101	Any dimension not as specified (see table I).	Visual
102	Materials (see 3.2).	Visual
Minor		
201	Color not as specified (see 3.6).	
202	Workmanship not as specified (see 3.7).	Visual
203	Marking not as specified (see 3.5).	Visual
204	Finish missing, blistered, peeled, chipped, scratched, or foreign matter imbedded (see 3.6).	Visual Visual
205	Bolts, screws, nuts, and washers not plated as specified (see 3.2.1).	Visual

TABLE IV. Classification of inspections.

Title	Requirement	Inspection	First article	Quality conformance test
Static load	3.4.1	4.4.2.2	X	X
Shock load	3.4.2	4.4.2.1	X	X

4.3 Conformance inspection. Conformance inspection shall include the examinations of 4.4 and the tests of 4.4.2.1, 4.4.2.2 and 4.5 (see table IV).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

4.4 Examination.

4.4.1 Sampling. Samples from an inspection lot for conformance inspection shall be selected as detailed in the contract. Any redesign or modification of the contractor's standard to comply with specific requirements shall receive particular attention for adequacy and suitability. This element of inspection shall encompass all visual examinations and dimensional measurement of requirements of 3.3, 3.3.1, 3.3.2, 3.3.3, 3.5, 3.6, 3.7 as listed in table III. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection.

4.4.2 Test conditions.

4.4.2.1 Shock load. The truck shall be placed on a raised platform 25 mm (1 in) high. A load of 94 kg (204 lbs) shall be placed on each of the three shelves of the truck. The total weight of 283 kg (624 lbs) shall be equally distributed over the three shelves. The loaded truck shall then be rolled completely off the raised platform at a minimum speed of 2 miles per hour (mph) onto a level concrete surface. This test shall be repeated 10 times with the direction of travel perpendicular to the step, and 10 times with the direction of travel at a 45 degree angle to the step. After completion of the test, the load shall be removed and the truck shall be inspected. Nonconformance to 3.4.2 shall constitute failure of this test.

4.4.2.2 Static load. The ladder legs shall be extended to a maximum spread and placed on a dry, level, concrete surface. A load of 181 kg (400 lbs) shall be placed centrally on one step of the ladder over an area of 127 x 152 mm (5 x 6 in) maximum. The ladder shall remain under load for a period of 10 minutes, after which the load shall be removed and the ladder inspected. The test shall be repeated for each individual step. Nonconformance to 3.4.1 shall constitute failure of this test.

4.5 Method of inspection.

4.5.1 Materials. Conformance to 3.2 shall be determined by inspection of contractor records providing proof of certification that materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.5.2 Defects. Conformance to 3.3, 3.3.1, 3.3.2, 3.3.3, 3.5, 3.6 and 3.7 shall be determined by examination for the defects listed in table IV. Examination shall be visual, tactile, or by measurement with Standard Inspection Equipment (SIE).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

4.5.3 Test conditions. Unless otherwise specified (see 6.2), ambient air temperature shall be 23 ± 10 °C (73 ± 18 °F), barometric pressure shall be 725 +50 -75 mm mercury (Hg) (28.5 +2, -3 inches Hg), and relative humidity shall be 50 ± 30 percent.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

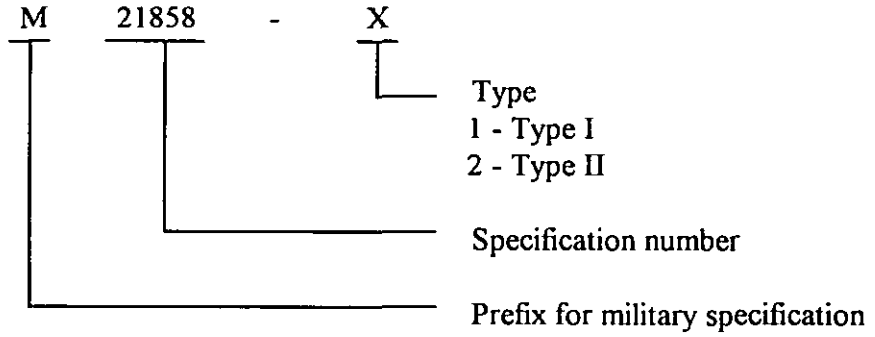
6.1 Intended use. The trucks covered by this specification are primarily intended for use by stock pickers in bin areas of military warehouses and depot facilities to transport small miscellaneous items. Type II truck (with ladder) adds an additional 762 mm (30 in) for operator to reach high bins.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2 and 2.3).
c. Type, PIN, and quantity of trucks required.
d. If first article samples are not required (see 3.1).
e. Color, of other than gloss yellow (see 3.6).
f. Test conditions if other than as specified (see 4.5.3).
g. Packaging requirements (see 5.1).

Source: https://assist.dla.mil -- Downloaded: 202610-19T10:31Z
Check the source to verify that this is the current version before use.

6.3 Part identification number (PIN). The trucks should use the following PIN structure:



6.4 Subject term (key word) listing.

Cart
Dolly
Ladder
Step

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Army - AT
Navy - SA
Air Force - 99

Review Activities:
Navy - MC
Air Force - 84
DLA - GS

Preparing Activity:
Army - AT

(Project 3920-0009)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. FSC 3920

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-PRF-21858E	960120

3. DOCUMENT TITLE
TRUCK, HAND, SHELF, STOCK PICKER (METRIC)

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (*Last, First, Middle Initial*)	b. ORGANIZATION	
c. ADDRESS (*Include Zip Code*)	d. TELEPHONE (*Include Area Code*) (1) Commercial (2) AUTOVON (*If applicable*)	7. DATE SUBMITTED (*YYMMDD*)

8. PREPARING ACTIVITY	
a. NAME	b. TELEPHONE (*Include Area Code*) (1) Commercial: (810) 574-8745 (2) AUTOVON: 786-8745
c. ADDRESS (*Include Zip Code*) Commander U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2020-10-19T10:31Z
Check the source to verify that this is the current version before use.